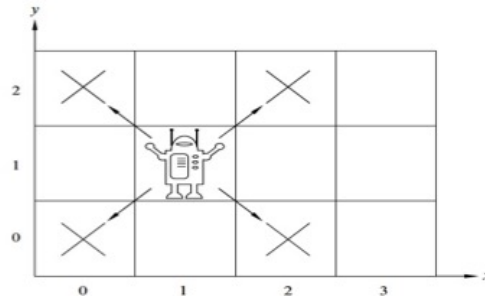


Structural induction example robot grid



Theorem: A robot on an infinite 2-dimensional integer grid starts at $(0,0)$ and at each step moves to diagonally adjacent grid point. This robot can / cannot (*circle one*) reach $(1,0)$.

Definition The set of positions the robot can visit Pos is defined by:

Basis Step: $(0,0) \in Pos$

Recursive Step: If $(x,y) \in Pos$, then

are also in Pos

Example elements of Pos are:

Lemma: $\forall (x,y) \in Pos$ ($x+y$ is an even integer)

Why are we calling this a lemma?

Proof of theorem using lemma: To show is $(1,0) \notin Pos$. Rewriting the lemma to explicitly restrict the domain of the universal, we have $\forall (x,y) ((x,y) \in Pos \rightarrow (x+y \text{ is an even integer}))$. Since the universal is true, $((1,0) \in Pos \rightarrow (1+0 \text{ is an even integer}))$ is a true statement. Evaluating the conclusion of this conditional statement: By definition of long division, since $1 = 0 \cdot 2 + 1$ (where $0 \in \mathbb{Z}$ and $1 \in \mathbb{Z}$ and $0 \leq 1 < 2$ mean that 0 is the quotient and 1 is the remainder), $1 \bmod 2 = 1$ which is not 0 so the conclusion is false. A true conditional with a false conclusion must have a false hypothesis: $(1,0) \notin Pos$, QED. $\square$

Proof of lemma by structural induction:

Basis Step:

Recursive Step: Consider arbitrary $(x, y) \in Pos$. To show is:

$(x + y \text{ is an even integer}) \rightarrow (\text{sum of coordinates of next position is even integer})$

Assume as **the induction hypothesis, IH** that: